PH4505 Computational Physics

Project Report

Monte Carlo Photon Transport in Multi-layered Biological Tissues

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Abstract

In this project, we implemented a Monte Carlo simulation for photon transport in multi-layered biological tissues based on the scattering and absorption coefficients. Photons are launched one at a time. Scattering directions and angles are sampled by a random walk using Henyey-Greenstein's function with the anisotropy factor. Absorption is simulated by assigning a weight to the photon and terminated with a Roulette method. Interactions at the boundaries are accounted for, such as total internal reflection, refraction and internal reflectance calculations. The weight that are lost through absorption and through escaping the tissue are recorded along with their trajectories, they will be used for calculations of absorption, reflectance and transmittance of the tissue layers. We demonstrated the program's simulation with a simple skin-skull-brain system and visualised the results with multiple plots. This project is important for calculating quantities and observables to be used in medical applications such as biomedical imaging and radiation therapy.

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Chapter 1

Introduction

1.1 Background

Biomedical imaging and radiotherapy have always been the main focus of medical physics. Advancements in imaging allows diagnosis to be made with a high confidence level as well as to identify tumours effectively [1], while precise calculations of radiotherapy's dosage and angle of incidence are crucial in killing tumours with minimal exposure risks to the patient.

But such calculations are non-trivial as it involves myriads of particles bombarding on biological tissues with different properties. Hence, computer simulations are commonly used to calculate quantities such as dosage absorption or backscattering reflectance, with the most commonly used method being the Monte Carlo method [2]. There are various programs that allows the simulation of particle transports using Monte Carlo methods, one example is GEANT4 [3], which is widely used around the world in the medical physics field as well as for particle accelerators.

However, there seems to be little or no programs that are used mainly for simple photon transports in tissues and medical physicists most often write their own programs to fulfil a niche application using the

many well-studied techniques [4]. Therefore, we wish to write a simple but general program to simulate photon transports in multi-layered tissues using one of the widely used techniques described by Lux et al. [5] and implemented in C by Wang et al. [6]. The objective of our program is to be able to measure various quantities that might be useful for biomedical imaging and radiotherapy. It must be general enough to be able to simulate any number and any type of tissue layers.

1.2 Theoretical Framework of Photon Interactions

The main interactions that a photon experiences when it passes through a medium are - photoelectric effect, Compton scattering, Rayleigh scattering, triplet production, coherent and incoherent scattering, nuclear photo-effect and pair production. Simulating for all types of interactions are computationally extensive. Thankfully, we can categorise interactions into two main areas - scattering and absorption, and these two interactions can be described simply with their corresponding coefficients - μ_s and μ_a.

These coefficients are known as the scattering and absorption coefficients, and are defined as the probability of scattering and absorption per unit pathlength respectively. Thus, all of the various interactions are summed together into these two coefficients and there are no shortage of literatures on the values of these two coefficients for different biological tissues, mostly obtained experimentally [7][8]. From the definition of these two coefficients, simulation of scattering and absorption can be done with Monte Carlo probabilistic random sampling.

These two coefficients will be the main variables differentiating between different layers, other differences include the refractive index and the anisotropy factor. The anisotropy factor will be discussed in the later chapter when calculating the deflection angles. By using these two coefficients, we are able to generalise any type of tissues as long as we can obtain their coefficients from literatures or databases. Our simulation will consists of multi-layered flat and homogenous tissues. This is a close enough approximation at a small scale.

1.3 Overview

In this report, we will first go through a lengthy description of the methods used in simulating photon transport, starting from explaining our choice of coordinates, to the scattering and absorption simulation and followed by the interactions at the boundaries of the tissue layers. Next, we will explain how the quantities such as reflectance, transmittance and absorption are measured, and finally, to conclude the methodology chapter, we will give a brief description of the code structure.

In the next chapter, we will present the results of a simple simulation of a real biological system, showing the various plots and results obtained. There will not be a detailed discussion on the explanation of the results. Rather, we will focus more on validating whether our results agree with real world experiments and behaviours as the objective of this simulation is not to solve a certain transport problem, but to provide a general program that can be used for various biological tissues.

Lastly, we will be identifying some limitations of the program, as well as demonstrating further improvements to the program that are being or can be worked on, such as a working GUI, wavelength dependence for the coefficients and a scaling method to reduce computational time. This is followed by a brief summary which concludes the report.

Chapter 2

Methodology and Implementation

In this chapter, we will discuss every step of the photon transport, from the launch of the photon into the medium to its termination or exiting out of the medium. We will only discuss the methods and make little references to how they are implemented in Python as it is basic without any complex algorithm. This chapter consists of three main sections, simulating the photon propagation and the measurement of the various quantities, which mimics the the two main parts of the implementation of the program, and lastly an overview of the code structure.

2.1 Simulating Photon Propagation

2.1.1 Coordinates

First, we have to choose a coordinate to define the position of the photon and its direction of propagation. A cylindrical coordinate [9] might be a natural choice due to cylindrical symmetry of the propagation, and allow us to calculate quantities that are in cylindrical coordinates easily such as beam spread. However we choose to opt for a Cartesian coordinate [10] as it is much simpler to implement as well as being a more widely used coordinate. We will define the following coordinates in Figure 2.1.

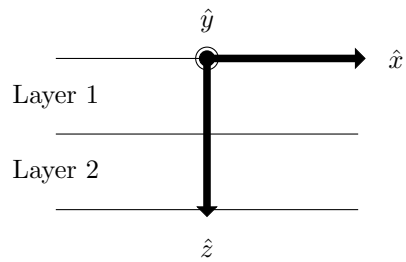

Figure 2.1: Coordinate System

Where $\hat{y}$ is pointing out of the paper. Here, we inverted $\hat{z}$ for downwards to be positive. However, when plotting the trajectories or any other visualisation of the results, we made sure to multiply the $\hat{z}$ coordinates by -1, such that the photons will be visualised as propagating downwards instead of upwards in the plots as positive $\hat{z}$ result in an upward plot in Python.

Next, as we are using Cartesian coordinates, defining the directions of propagation is simple with the directional cosines, which is one of the reasons we have chosen Cartesian coordinate.

$$\mu_x = \vec{r} \cdot \hat{x}$$

$$\mu_y = \vec{r} \cdot \hat{y}$$

$$\mu_z = \vec{r} \cdot \hat{z}$$

The directional cosines are the cosine of the angles between the propagation vector $\vec{r}$ and the coordinate unit vectors $\hat{x}, \hat{y}, \hat{z}$. Hence, they can be represented by their dot products. Therefore, the update to the position of a photon is simply:

$$x \leftarrow x_{\text{old}} + \mu_x \cdot s_i$$

$$y \leftarrow y_{\text{old}} + \mu_y \cdot s_i$$

$$z \leftarrow z_{\text{old}} + \mu_z \cdot s_i \tag{2.1}$$

Here, we used an arrow to indicate an assignment in Python. s_i is the step size of the propagation. We included a subscript i to indicate that this is the step size for the ith step as the step size is not fixed and

varies at every step. We will discuss the calculation of the step size in the next section. Note that the directional cosines happens to use the same notation μ as the scattering and absorption coefficients μ_s and μ_a, they describe different things and their same notations are simply a coincidence in the way directional cosines and the coefficients are described.

2.1.2 Scattering - Random Walk

Photon scatterings are simulated through a series of random walk, where at each interaction, we will update to a new set of directional cosines to calculate the new positions with equations (2.1). First, the step size which is the distance between interactions or the photon's free path must be determined. This is simple as the scattering and absorption coefficients are defined as the probability of scattering and absorption per unit length. Hence, if we define the total interaction coefficient μ_t (linear attenuation coefficient in some literatures [11]):

$$\mu_t = \mu_s + \mu_a = N\sigma_{tot}$$

$$\sigma_{tot} = \sigma_{pe} + \sigma_{coh} + \sigma_{incoh} + \sigma_{pair} + \sigma_{trip} + \sigma_{phn}$$

Where σ is the cross sections of different interactions. N is the number of target entities per unit volume and is defined as $\frac{N_A}{A}\rho$. Where N_A is the Avogadro's number, A is the relative atomic mass of the target element and ρ is its density. However, we will not concern ourselves with N and σ here as the values of μ_s and μ_a and thus μ_t for different tissues are well-defined in various literatures.

μ_t is the probability of interaction per unit length. Hence, for a length interval s_i to $s_i + ds_i$:

$$\mu_t = \frac{-dP(s \geq s_i)}{P(s \geq s_i)ds_i}$$

$$\implies d(\ln(P(s \geq s_i))) = -\mu_t ds_i$$

Integrating from 0 to s_1,

$$P(s \geq s_1) = e^{-\mu_t s_1}$$

$$\implies P(s < s_1) = 1 - e^{-\mu_t s_1}$$

Which is a cumulative probability distribution function. Assuming a cumulative probability distribution function normalised from x_0 to x_1, it is monotonically increasing from 0 to 1, hence in the range x_0 to x where $x_0 < x < x_1$:

$$c(x) = \int_{x_0}^{x} dx' p(x')$$

$$\implies x = c^{-1}(R)$$

Where R is a uniformly distributed random number between 0 and 1. Similarly, we can use R to sample our probability distribution function and get:

$$s_i = \frac{-\ln(1-R)}{\mu_t}$$

$$\implies s_i = \frac{-\ln(R)}{\mu_t} \tag{2.2}$$

Where we substituted R as $1 - R$. Equation (2.2) will be used to calculate the step size.

Next, we have to determine the direction of scattering at every step. The deflection angle $\theta \in [0, \pi)$ and the azimuthal angle $\phi \in [0, 2\pi)$ will also be sampled statistically using random number. We will use the scattering function defined by Henyey and Greenstein in 1941 [12]. It was used for galactic scattering, but have been found to be equally effective for photon scattering experimentally [13]. The Henyey-Greenstein function describe the probability distribution of $\cos\theta$:

$$P(\cos\theta) = \frac{1-g^2}{2(1+g^2-2g\cos\theta)^{3/2}}$$

$$\cos\theta = \begin{cases} \frac{1}{2g}\left[1+g^2-\left[\frac{1-g^2}{1-g+2gR}\right]\right] & \text{if } g \neq 0 \\ 2R-1 & \text{if } g = 0 \end{cases} \tag{2.3}$$

Equation (2.3) will be used to determine the deflection angle θ. g is the anisotropy. As for the azimuthal angle ϕ, it is sampled uniformly due to cylindrical symmetry. Therefore, $\phi = 2\pi R$, where R is another uniformly distributed random number from 0 to 1.

Having the deflection and the azimuthal angles, we can now update the direction cosines to the scattering direction:

$$\begin{aligned}
\mu_x &\leftarrow \frac{\sin\theta}{\sqrt{1-\mu_z^2}}(\mu_x\mu_y\cos\phi - \mu_y\sin\phi) + \mu_x\cos\theta \\
\mu_y &\leftarrow \frac{\sin\theta}{\sqrt{1-\mu_z^2}}(\mu_y\mu_z\cos\phi + \mu_x\sin\phi) + \mu_y\cos\theta \\
\mu_z &\leftarrow -\sin\theta\cos\phi\sqrt{1-\mu_z^2} + \mu_z\cos\theta
\end{aligned} \tag{2.4}$$

These trigonometric calculations will take up most of the computational time of the program. For photons that have $|\mu_z| > 0.99999$, we can make the approximation of:

$$\begin{aligned}
\mu_x &\leftarrow \sin\theta\cos\phi \\
\mu_y &\leftarrow \sin\theta\sin\phi \\
\mu_z &\leftarrow \mathrm{SIGN}(\mu_z)\cos\theta
\end{aligned} \tag{2.5}$$

Now, having sampled the directional cosines and the step size statistically with random numbers, we can update the position with equation (2.1). This process of updating the directional cosines and the position of the photon will be carried out continuously until the the photon exits the layers of tissues, or when it is fully absorbed. We will talk about absorption in the next section and the interactions with the boundary of the tissues in the later sections.

2.1.3 Absorption and Photon Termination

If there are no absorption, the photon will propagate endlessly in the tissues or until it exits the tissue layers where we will terminate it with a condition statement. Absorptions are simulated through the attenuation of the weight of the photon. At every step the photon took, we will calculate the the amount of weight that

will be absorbed using the absorption coefficient μ_a:

$$\Delta W = W\frac{\mu_a}{\mu_t}$$

$$\therefore W \leftarrow W - \Delta W \tag{2.6}$$

The weight of a photon will also be used for the measurement of the absorption, reflectance and transmittance which will be discussed later. Hence, once the photon reach a certain threshold weight, we could simply terminate the photon as it would not contribute to our results significantly due to the small remaining weight. However, the photons must be terminated with conservation of energy and the distribution of photon deposition in mind [5]. We will use the roulette method, frequently credited to von Neumann [14], to give photons beneath the threshold weight a 1 in m chance of surviving termination with a weight of mW:

$$W \leftarrow \begin{cases} mW & \text{if } R \leq 1/m \\ 0 & \text{if } R > 1/m \end{cases} \tag{2.7}$$

This roulette method is frequently used with the particle splitting method. Such that when the particle is in the area of interest, there will be some probability to split a particle into two, and when the particle is not in the area of interest, it will be terminated with the roulette method. However, we will not be using the particle splitting method as all parts of the tissue layers are our area of interest, and we wish to have the number of photons fixed for easier calculations. Though, using both methods together can reduce the variance of the results more [15].

2.1.4 Boundary Interactions

Next, we need to simulate photon interactions at the boundaries. There are three main types of boundary interactions. First, when the photon enters the tissue layers, there will be specular reflectance deducting the weight of the photon. Specular reflectance is calculated simply with:

$$R_{sp} = \frac{(n_1 - n_2)^2}{(n_1 + n_2)^2} \tag{2.8}$$

Where n_1 and n_2 are the refractive index of the outside boundary (air) and the first layer of the tissue respectively. Specular reflectance are the reflectance from the photons before they interact with the medium.

There are cases whereby the photon can pass through the first layer without any interaction, this happens when the first layer is very thin, such as the skin layer. Then, we need to account for the refractive index of the second layer in the calculation of specular reflectance:

$$R_{sp} = r_1 + \frac{(1-r_1)^2 r_2}{1 - r_1 r_2}, \tag{2.9}$$

$$\text{where } r_1 = \frac{(n_1 - n_2)^2}{(n_1 + n_2)^2},$$

$$r_2 = \frac{(n_3 - n_2)^2}{(n_3 + n_2)^2}$$

r_1 and r_2 are the Fresnel reflectance. n_3 is the refractive index of the second layer. This is illustrated as:

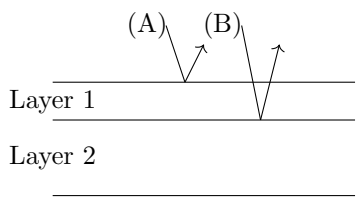

Figure 2.2: Specular Reflectance - (A), we will use equation (2.8). (B), equation (2.9).

The second type of boundary interaction is the interactions between two layers inside the tissue - the tissue-tissue boundary. Due to the difference in refractive index, we would expect to calculate for internal reflectance and the probability of reflecting or transmitting. But first, we have to check if the photon will hit the boundary. This is done by calculating the distance to the boundary and comparing it with the step size.

$$s_d = \begin{cases} \frac{z - z_0}{-\mu_z} & \text{if } \mu_z < 0 \\ \frac{z_1 - z}{\mu_z} & \text{if } \mu_z > 0 \end{cases} \tag{2.10}$$

Where s_d is the distance to the boundary. z is the z position of the photon, z_0 is the depth of the previous layer and z_1 is the depth of the next layer. There are no cases for $\mu_z = 0$ as in this case, the photon will be travelling parallel to the boundary and hence would not hit it. If the step size s_i is larger than s_d, this

means that the photon will hit the boundary, so we will move the photon to the boundary first with:

$$x \leftarrow x_{\text{old}} + \mu_x \cdot s_d$$

$$y \leftarrow y_{\text{old}} + \mu_y \cdot s_d$$

$$z \leftarrow z_{\text{old}} + \mu_z \cdot s_d$$

And the step size is thus updated to $s_i \leftarrow s_i - s_d$. Next, we shall check for total internal reflection by calculating the angle of incidence,

$$\alpha_i = \cos^{-1} |\mu_z|$$

and hence total internal reflection occurs when:

$$\frac{n_i \sin \alpha_i}{n_t} > 1 \tag{2.11}$$

In this case, we simply flip the z directional cosine: $\mu_z \leftarrow -1 \times \mu_z$. However, if there are no total internal reflection, we will have to calculate the internal reflectance. Using Snell's law,

$$n_i \sin \alpha_i = n_t \sin \alpha_t$$

The internal reflectance can be calculated using Fresnel's formulas [16]:

$$R(\alpha_i) = \frac{1}{2}\left[\frac{\sin^2(\alpha_i - \alpha_t)}{\sin^2(\alpha_i + \alpha_t)} + \frac{\tan^2(\alpha_i - \alpha_t)}{\tan^2(\alpha_i + \alpha_t)}\right] \tag{2.12}$$

The internal reflectance is defined as the average reflectance of the two orthogonal polarisation directions. This internal reflectance is also the probability for the photon to internally reflect, which we will again sample with a uniformly distributed random number from 0 to 1. If the photon does not internally reflect, it will transmit and we will simulate refraction by:

$$\begin{aligned} \mu_x &\leftarrow \mu_x \times \frac{n_i}{n_t} \\ \mu_y &\leftarrow \mu_y \times \frac{n_i}{n_t} \\ \mu_z &\leftarrow SIGN(\mu_z) \times \cos \alpha_t \end{aligned} \tag{2.13}$$

If it does reflect, similar to total internal reflection, we flip $\mu_z \leftarrow -1 \times \mu_z$.

It is suggested that instead of this all or nothing approach whereby the photon either fully reflect or transmit, we can instead allow the photon to have a partial reflection whereby a fraction of the photon's weight is transmitted and the others reflected [6]. However, doing so for the tissue-tissue boundary will require us to split a photon into two, which we tries to avoid as mentioned previously when talking about the Roulette method. But, we can implement this splitting approach at the air boundaries of the tissues as we do not have to further simulate any photons that exited the tissue into the surrounding air medium, which brings us to the third main boundary interaction - air-tissue boundary.

There are only two of such air-tissue boundaries, the top and the bottom. For both, we will use the same method as the tissue-tissue boundary. The only difference is that every time the photon hits the boundary, it will be guaranteed to be reflected with its weight reduced unless total internal reflection occurs which the photon will not lose any weight. The fraction of weight lost is:

$$\begin{aligned} &\Delta W = W(1 - R(\alpha_i)) \\ &\therefore W \leftarrow W - \Delta W \quad \text{or} \quad W \leftarrow WR(\alpha_i) \end{aligned} \tag{2.14}$$

Hence, we only need to account for the weight reduction when the photon hits the air-tissue boundary as the photon carrying the lost weight out of the tissue layers need not be simulated. The photon weights that escape from the top boundary are known as diffuse reflectance, whereas those that escape from the bottom, are known as diffuse transmittance which we will talk more about in the next section.

Figure 2.3 summarises all the boundary interactions. (A) - specular reflectance. (B) - tissue-tissue boundary interaction whereby the internal reflectance is the probability of reflecting. (C) - diffuse transmittance whereby some of the photon weight is lost with the remaining reflecting. (D) - diffuse reflectance, similar to (C) but the weight lost is at the top layer.

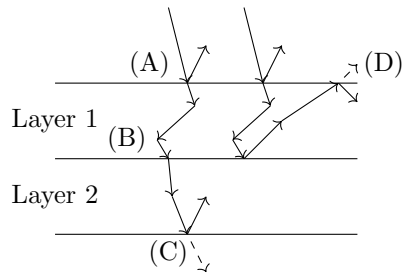

Figure 2.3: Boundary Interactions

2.2 Measured Quantities

We will measure 4 quantities from our simulation - specular reflectance, diffuse reflectance, diffuse transmittance and absorption. These quantities are chosen because they are useful in the field of biomedical diagnostic imaging and photon therapy. Each run will consists of simulating the random walks of 10000 photons, for a total of 10 bins.

2.2.1 Reflectance and Transmittance

It's not the end of the story for the escaped or lost photon weight at the top and bottom of the tissue. They will be recorded along with their x and y coordinates of exit. z coordinate is not necessary as it is obviously 0 for the top boundary and the total depth for the bottom boundary.

The collection of these data are done in the class file, whereas analysis of these data are done in the main file. The total diffuse reflectance and transmittance are simply calculated by summing up all the lost weight at the top and bottom layer respectively. This sum is then divided by the total number of photons N:

$$\begin{aligned} R_{tot} &= \frac{\sum \Delta W_{top}}{N} \\ T_{tot} &= \frac{\sum \Delta W_{bottom}}{N} \end{aligned} \tag{2.15}$$

This gives us the percentage of diffuse reflectance and transmittance as the weight of each of the N photons is 1. The x and y coordinates are used to determine the radial spread of the diffuse reflectance and trans-

mittance, a heatmap can be plotted to visualise the data. The total diffuse reflectance and transmittance are useful in determining the amount of backscattering or transmittance for purpose of diagnostic imaging.

As for the specular reflectance, similarly, it is summed over and divided by N. There are no radial spread for the specular reflectance as it only happens at the point of entry, unlike diffused reflectance.

$$R_{sp_{tot}} = \frac{\sum R_{sp}}{N} \tag{2.16}$$

Specular reflectance is useful for determining the amount of reflected and lost photon beam.

2.2.2 Absorption

Similarly for absorption, the x, y, z coordinates are recorded along with the weight absorbed. Again, the total absorption is:

$$A_{tot} = \frac{\sum \Delta W_{absorbed}}{N} \tag{2.17}$$

The amount of absorption is very important for dose calculations in radiotherapy. Hence, we also wish to determine the radial spread of the absorption as well to plot a heatmap to visualise the absorption. Besides the reflectance, transmittance and absorption, we also recorded the trajectories of the photons for visualisation as well as for the scaling method.

2.3 Code Structure

The purpose of this section is to mention some subtleties in the implementation of the program. The full code can be referred to in the Appendix. First, all the aforementioned calculations in simulating random walk and absorption are implemented in a class file - *build.py*. The Layer class will store all the constants and coefficients required for each layer such as:

1. Refractive Index - ref_index

2. Scattering and Absorption Coefficients - sct_coeff, abs_coeff

3. Anisotropy Factor - g

4. Depth - depth

5. Diffuse Reflectance - diff_reflect

6. Diffuse Transmittance - diff_trans

7. Specular Reflectance - spec_reflect

8. Absorption - deposit

Here we note that the depth d is not the thickness of the layer, if the thickness of the first and second layers are both 0.5cm, their depths are 0.5cm and 1cm respectively. diff_reflect, diff_trans and deposit are initialised as an empty list whereas spec_reflect is initialised as 0. The layer class also has two methods - plotLayer2D and plotLayer3D, which plots the boundaries of the layers. The last method is reset which reset all values to the initialised values. This will be used if we wish to remove all the stored data of diff_reflect, diff_trans, spec_reflect and deposit.

Next, for the Photon class, it has the variables:

1. Position - pos

2. Directional Cosines - mu

3. Weight - weight

4. Weight Threshold - threshold

5. m value for Roulette Method - roulette_m

6. Dead or Alive Status - dead

7. Stored Trajectories - traj

The Photon class contains 8 methods including the reset method. The methods are to implement the scattering and absorption described in section 2.1. 5 of them are specular reflectance, directional cosines update, absorption, check if dead and check for boundary interactions. The startScatter method simply put these 5 methods together to describe a single scattering step. The last method, startWalk simply loop the startScatter method until dead == 1. At each scattering step, the photon class will check the layer it is in so as to use the correct coefficients for its corresponding layer. The measured data are then stored into the variables in the Layer class that the photon is in. The main file - *main.py* simply contains the functions to run the simulation, visualise the data and to do the calculations for analysis of the results.

Chapter 3

Results and Discussions

We launched 10000 photons, with initial positions of [0, 0, 0] and initial directional cosines of [0, 0, 1] which means the photons are launched in the z direction into the tissue layers at the origin. The simulated layers were: skin - skull - brain, with the following coefficients:

Layers	n	μ_a	μ_s	d	g
Skin	1.359	0.26	48.9	0.1	0.75
Skull	1.555	0.07	9.5	1.8	0.925
Brain	1.44	3.84	40.8	5	0.875

Table 3.1: Coefficients for skin - skull - brain layers[7][8]

The depth of the brain is arbitrary as long as it is large, as we do not expect the photon beam to transmit out of the brain to the other side of the head. The coefficients obtained here are for a wavelength of 500nm. We will talk more about the wavelength dependence of the coefficients in the next chapter. The units of the coefficients are [cm^{-1}] and for the depth, [cm]. The coefficients and the depth must have the same unit scale. For the photons, the threshold weight is set at 1e-4 and the roulette m is set to be 10. These choices are based on the choices of Wang et al. [6].

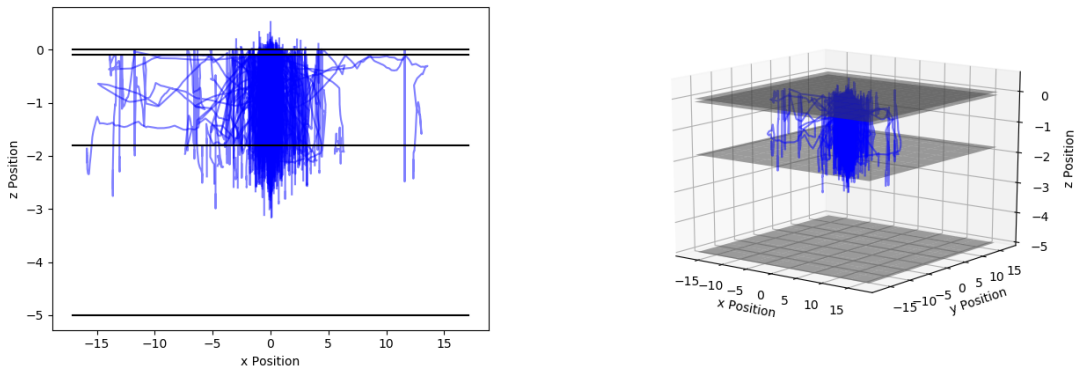

Figure 3.1: 2D and 3D Trajectories of 500 Photons

Figure 3.1 shows the trajectories of 500 photons. We recorded the trajectories for all 10000 photons, but plotting too many trajectories might slow the program, especially for the 3D plot. The diffused reflectance (escaped photons) are not visualised as technically no photons exit the layers, we only store their weight loss and reflect the photon. However, there are still some trajectories that come out from the top. These are stray photons that happens to scatter backwards.

The photons are unable to penetrate deep into the brain layer. This is because the absorption coefficient of the brain layer is high and thus most of the photons were absorbed upon entering the brain layer. We visualise the absorption in Figure 3.2.

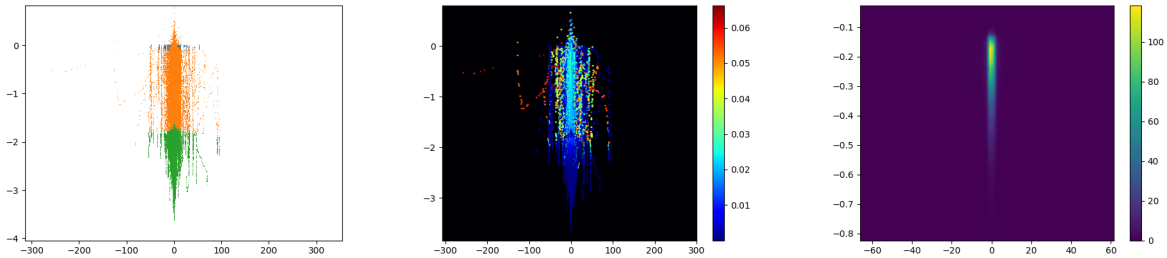

Figure 3.2: Density, hexbin and 2Dhistogram heatmap plots of absorption

The plots of Figure 3.2 consists of all 10000 photons, hence the spread might look different from the 500 photons trajectories. The data plotted are the layer classes' deposit variable, which contains the coordinates and amount of weight absorbed. The first plot is a pixel scatter plot, from it we can see the density of the area of absorption. Since it is a pixel plot, area with a lot of absorption will appear to be filled up. To account for the amount absorbed or the weight of the absorption, we decided to use the built-in function of matplotlib - 'hexbin' and '2Dhistogram'. The centre plot showing the hexbin plot turns out to not be a good visualisation as it appears messy, but from it we can roughly see that majority of the absorption happens in the bone layer. The third plot, using the 2Dhistogram and bilinear interpolation resulted in a smoother plot, showing how the absorption is maximum at the centre and decays as it goes deeper into the tissues.
As for the other quantities:

$$R_{tot} \approx 0.32 \pm 0.02$$

$$R_{sp_{tot}} \approx 0.02316$$

The value of the specular reflectance is fixed as it depends on the refractive indices which are also fixed. As for the diffuse reflectance, a value of 0.32 means that a total of around 32% of the photon weight were backscattered. The error can be calculated from repeated runs or binning. Figure 3.3 shows the radial spread of the diffuse reflectance. The number of backscattered photons as well as their weights are maximum near the origin, thus the heatmap is concentrated at a single point.

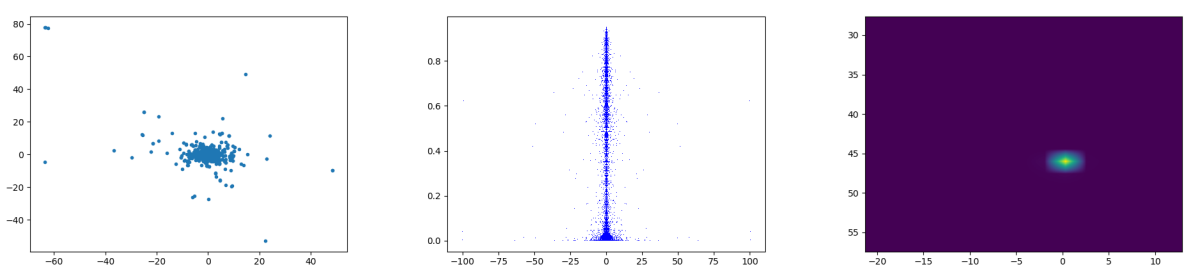

Figure 3.3: Top view scatter, weight against radial and heatmap

We would like to note that these results serve only as a proof of concept for scoring physical quantities from our simulation. Although our data analysis and plotting are done in the main file - *main.py*, the main focus of this project is to write a general program for simulating photon transports for any type of systems, and have any interested quantities or observables scored in a database for analysis by the user. In this chapter, we measured the absorption and reflectance and visualised their spread which are important for dosage calculations, such as in the application of killing a tumour in the brain. But there are also various other quantities that can be scored depending on the application. For example, the angle of escape or diffuse transmittance might be a necessary data for x-ray or other tomographic imaging. (Elaine simulated x-ray transmission through a hand)

Chapter 4

Conclusion

4.1 Limitations and Further Improvements

In the simulation, photons are treated as particles without wavelike properties. Interference and polarisation are not accounted for. Photons also do not interact with each other as only one photon were launched at a time. Some interactions such as pair production produce secondary cascades that might interact with the photon beam, this is also not accounted for, though it is simple to implement but requires addition coefficients for electrons and positrons.

There are some improvements that we plan to implement. First, wavelength dependence - the coefficients μ_s, μ_a, g, n all depends on wavelength. This means that different wavelengths will result in different behaviours. For example,

$$\mu_s(\lambda) = a\left(\frac{\lambda}{500nm}\right)^{-b}$$

where $a = \mu_s(\lambda = 500nm)$ and b is the scattering power. Although the wavelength dependence is simply a matter of implementing an equation, we decided not to implement it in our program as implementing the wavelength dependence simply for μ_s alone would include two additional user inputs - the wavelength of the input μ_s and the wavelength the user wants. Furthermore, the various coefficients we obtained from various

literatures were already given in various different wavelengths, the user can simply enter the desired wavelength's coefficients. However, wavelength dependence can still be implemented if we make it user-friendly enough which brings us to the next point, the Graphics User Interface (GUI).

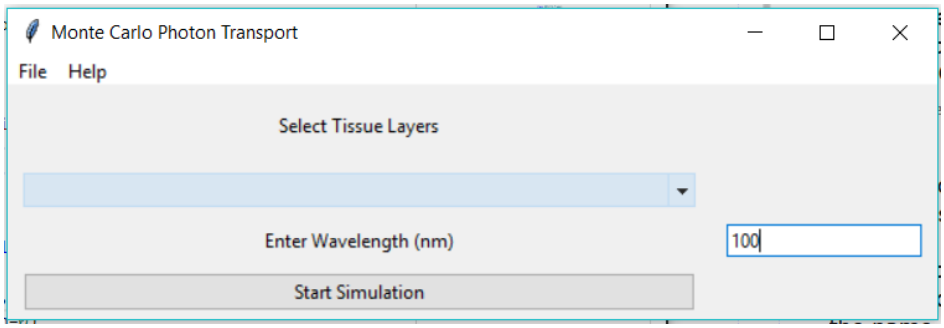

Figure 4.1: Prototype GUI

We have began working on a GUI which will allow us to implement a database of coefficients for various biological tissues, selectable from perhaps a drop-down menu. Users would only have to select the tissues for different layers and enter a wavelength or energy for the photon. However, due to time constraint, the GUI will only be done after the improvements to other parts of the program.

Next, we have also implemented a scaling method, whereby we are able to use the trajectories of a previous run to calculate the trajectories of a new run. Currently, we have done this for scaling of the total interaction coefficients μ_t. For example, for a run of $\mu_t/2$, we know that the step size would be twice of a run of μ_t. Therefore, we can use the trajectories for the run of μ_t to calculate the new trajectories for $\mu_t/2$ simply by multiply the step size by 2.

Figure 4.2 shows the original trajectories on the left, and the scaled trajectories on the right. By using the scaling method, we can save simulation time as calculating the scaled trajectories is significantly faster than rerunning the whole simulation. This scaling method can reduce computation time by 2 order of magnitudes with relative error of less than 4% [17][18].

Lastly, in our simulation, photons have a single initial position and a initial directional cosines. How-

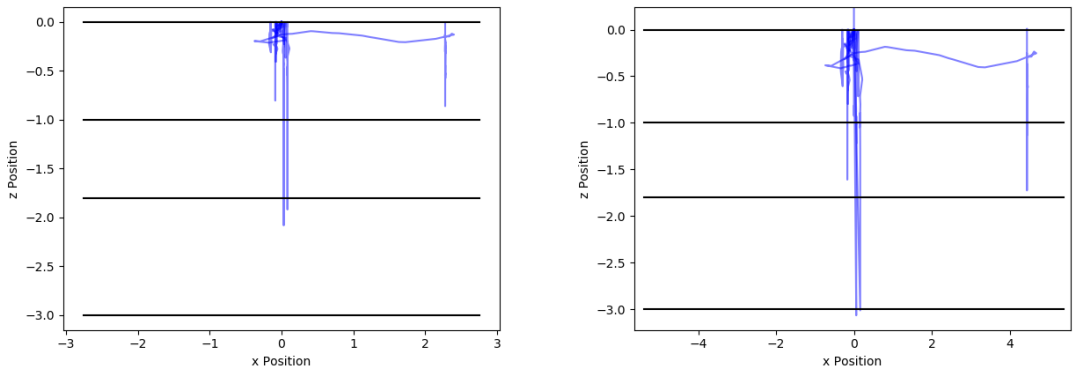

Figure 4.2: Scaled Trajectories

ever, this is not the case for real world photon beams. To improve the program, we can allow the user to input a beam radius in the GUI so that the photons follow a more realistic path instead of penetrating at a single position. This can be implemented by allowing the photons to vary its initial position, either as a sweep or from a probability distribution.

4.2 Summary

In conclusion, we were able to simulate photon transports in multi-layered biological tissues using Monte Carlo methods. Random walk was used to simulate scattering and weight reduction for absorption. The scattering step size was sampled from the total interaction coefficients, the deflection angle from the Henyey-Greenstein function using the anisotropy, and azimuthal angle sampled uniformly. A Roulette method was used for photon termination when they reach a threshold weight. The weight starts from 1 and is reduced at each step using the absorption coefficient. Other interactions such as boundaries reflection or transmission were also calculated using the refractive index of the involving layers. Total internal reflection were checked for, and the internal reflectance was used to determine reflecting or transmitting. Refraction were also accounted for when transmitting. If it is at the air-tissue boundaries, the photon will have some of its weight escaped while the remaining weight continues to propagate in the tissue.

The absorbed weight were recorded along with its position coordinates, the weight of the photons that escaped the tissue from both the top and the bottom are also recorded along with its coordinates as the diffuse reflectance and transmittance. We demonstrated the program's capability by simulating a skin-skull-brain systems and visualised the spread of the absorption and diffuse reflectance. We have identified some limitations such as not accounting for secondary cascades and wavelike properties. Finally we discussed and demonstrated some improvements that could be made to the program such as a GUI, scaling method and wavelength dependence.

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